数理分析方法

函数、对应与可数集

课件编号：103-ZSM-PPT　原始 PPT：35 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

证明像与交集的关系时，始终检查同一个 y 是否来自同一个 x；逆像的集合运算往往严格保等号。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。

第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。

第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。

第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。

极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。

存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 10, 2022

这一页怎么读

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结合原文理解：证明像与交集的关系时，始终检查同一个 y 是否来自同一个 x；逆像的集合运算往往严格保等号。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.2 第 2 页｜ Chapter 1: Mathematics

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

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本页重点：反身性是 xRx；传递性是 xRy 且 yRz 推出 xRz。完全性与反对称性不包含在预序定义内。

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2.3 第 3 页｜Functions

Functions

Let S and T be two sets; if with each element $x \in S$ is associated one and only one element $y \in T$, a function f from S to T is defined. f is also called a transformation of S into T. x is the variable, y is the image of x by f, or the transform of x by f, or the value of f at x, and one writes $y = f(x)$ (read y equals f of x), or $x \to f(x)$.

Example: Let A be the set of actions available to a certain agent. If the choice of a in A determines the amount of money (a positive integral number of cents) this agent receives, a function from A to $\mathbb{N}$ is defined.

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对应的完整证明：参见本讲义最后的「image」证明专题。

2.4 第 4 页｜ Functions

Functions

Consider in $S \times T$ the set of elements (x, y) for which $y = f(x)$. This subset of $S \times T$ is called the graph of the function f.

$$G(f) = \{(x, y) \in S \times T \mid y = f(x)\}.$$

Example: The graph of a function f from S to T is visualized as a set of points of the rectangle such that the vertical through an arbitrary point x of S intersects it at exactly one point.

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本页重点：函数图像是有序对集合。函数性要求每个 x 有且只有一个 y；对应图像允许同一个 x 对应多个 y。

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对应的完整证明：参见本讲义最后的「graph」证明专题。

2.5 第 5 页｜Functions

Functions

Let X be a subset of S; take the image $y = f(x)$ of each $x \in X$. The set of the images so obtained is a subset of T called the image of X and denoted $f(X)$.

$$f(X) = \{f(x) \in T \mid x \in X\}.$$

If $f(S)$ consists of one element of T, in other words, if all the elements of S have the same image in T, the function f is said to be constant. If $f(S) = T$, in other words, if each element of T is the image of some element of S, f is said to be a function from S onto T, i.e., function $f : S \to T$ is a surjective (onto) mapping.

For any $y \in T$, there exists $x \in S$ such that $f(x) = y$.

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2.6 第 6 页｜Functions

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\mathscr{X}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{X\in\mathscr{X}} X\right) = \bigcup_{X\in\mathscr{X}} f(X)$.

(2) $f\left(\bigcap_{X\in\mathscr{X}} X\right) \subseteq \bigcap_{X\in\mathscr{X}} f(X)$.

Remark: There exist a function f from a set S to a set T, and $\mathscr{X}$ is a collection of subsets of S, such that then $f\left(\bigcap_{X\in\mathscr{X}} X\right) \neq \bigcap_{X\in\mathscr{X}} f(X)$.

Example: Let $A = \{1,2\}$, $f(1) = f(2) = 0$, then $f(\{1\}) \cap f(\{2\}) = \{0\}$ while $\{1\} \cap \{2\} = \varnothing$.

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2.7 第 7 页｜Functions

Functions

Theorem 1. If f is a function from a set S to a set T, and if $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, then

(1) $f\left(\bigcup_{\lambda\in\Lambda} X_\lambda\right) = \bigcup_{\lambda\in\Lambda} f(X_\lambda)$.

(2) $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

Remark: There exist a function f from a set S to a set T, and $\{X_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of S, such that then $f\left(\bigcap_{\lambda\in\Lambda} X_\lambda\right) \neq \bigcap_{\lambda\in\Lambda} f(X_\lambda)$.

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2.8 第 8 页｜ Functions

Functions

Conversely, let Y be a subset of T; the set of $x \in S$ which have their images in Y is a subset of S called the inverse image of Y and denoted $f^{-1}(Y)$.

$$f^{-1}(Y) = \{x \in S \mid f(x) \in Y\}.$$

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对应的完整证明：参见本讲义最后的「image」证明专题。

2.9 第 9 页｜Functions

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\mathscr{Y}$ is a collection of subsets of T, then It is easy to prove that:

(1) $f^{-1}\left(\bigcup_{Y\in\mathscr{Y}} Y\right) = \bigcup_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(2) $f^{-1}\left(\bigcap_{Y\in\mathscr{Y}} Y\right) = \bigcap_{Y\in\mathscr{Y}} f^{-1}(Y)$.

(3) $f^{-1}(Y \setminus Y') = f^{-1}(Y) \setminus f^{-1}(Y')$ for $Y' \subseteq Y \subseteq T$.

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2.10 第 10 页｜Functions

Functions

Theorem 2. If f is a function from a set S to a set T, and if $\{Y_\lambda \mid \lambda \in \Lambda\}$ is a collection of subsets of T, then It is easy to prove that:

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(2) $f^{-1}\left(\bigcap_{\lambda\in\Lambda} Y_\lambda\right) = \bigcap_{\lambda\in\Lambda} f^{-1}(Y_\lambda)$.

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2.11 第 11 页｜ Functions

Functions

A $y \in T$ may be the image of several, or of one, or of no element of S. When for each $y \in T$ the set $f^{-1}(y)$ consists of exactly one element, in other words, when each $y \in T$ is associated with one and only one element of S, f is said to establish a one-to-one correspondence between S and T, i.e., function $f : S \to T$ is an injective (one-to-one) mapping.

For any $x_1 \in S$ and $x_2 \in S$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

For any $x_1 \in S$ and $x_2 \in S$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

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对应的完整证明：参见本讲义最后的「image」证明专题。

2.12 第 12 页｜ Functions

Functions

Let f be a function from a set F to a set T, and let G be a set containing F. A function g from G to T is said to be an extension of f to G if one has $f(x) = g(x)$, for every x in F.

Let f be a function from a set F to a set T, and let G be a set contained in F. A function g from G to T is said to be an restriction of f to G if one has $f(x) = g(x)$, for every x in G.

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2.13 第 13 页｜ Functions

Functions

Consider N sets $S_1, \cdots, S_n, \cdots, S_N$ and their product $\prod_{n=1}^{N} S_n$. The function which associates with the generic element (x_n) of $\prod_{n=1}^{N} S_n$, its nth coordinate x_n in S_n is called the nth projection (or the projection on S_n). The image of an element (resp. of a set) by a projection function is called the projection of that element (resp. of that set).

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2.14 第 14 页｜Correspondences

Correspondences

If with each element x in a given set S is associated a non-empty subset Y of a given set T, a function φ, from S to the set of subsets of T is defined. It is sometimes preferable to consider φ as a correspondence from S to T. One writes $Y = \varphi(x)$.

Example: (1) Let f be a function from a set S onto a set T; for every $y \in T$, $f^{-1}(y)$ is a non-empty subset of S, hence f^{-1} is a correspondence from T to S. (2) Let A be the set of a priori available actions of a social agent. Suppose that his environment is completely specified by an element e of a set E. His environment restricts his freedom of action, i.e., the element e determines the subset of A to which his choice is actually restricted. A correspondence from E to A is thus introduced.

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2.15 第 15 页｜Correspondences

Correspondences

Let S and T be two sets. If with each element x of S we associate a subset $\varphi(x)$ of T, we say that the correspondence $x \longrightarrow \varphi(x)$ is a mapping of S into T; the set $\varphi(x)$ is called the image of x under the mapping φ. The set $S^* = \{x \in S \mid \varphi(x) \neq \varnothing\}$ is called the domain (or set of definition) of φ and $T^* = \bigcup_{x \in S} \varphi(x)$ is called the range (or set of values) of φ; we also say that φ is defined on S^* and that it is a mapping of S^* onto T^*, i.e., correspondence $\varphi : S^* \to T^*$ is a surjective (onto) mapping.

If the mapping φ of S into T is such that the set $\varphi(x)$ always consists of a single element, we say that φ is a single-valued function or a single-valued mapping of S into T.

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对应的完整证明：参见本讲义最后的「image」证明专题。

2.16 第 16 页｜Correspondences A mapping is called semi-single-valued

Correspondences

A mapping is called semi-single-valued if

$$\varphi(x) \cap \varphi\left(x'\right) \neq \varnothing \Longrightarrow \varphi(x) = \varphi\left(x'\right).$$

Clearly a single-valued mapping is also semi-single-valued. A mapping is called injective (one-to-one) if

$$x \neq x' \Longrightarrow \varphi(x) \cap \varphi\left(x'\right) = \varnothing.$$

An injective mapping is evidently semi-single-valued.

If φ is a mapping of S into T and A is a non-empty subset of S, we write

$$\varphi(A) = \bigcup_{x \in A} \varphi(x).$$

If $A = \varnothing$, we write $\varphi(\varnothing) = \varnothing$. The set $\varphi(A)$ is called the image of A under the mapping φ; if we compare x to a source of light, making a shadow $\varphi(x)$ on a screen, then $\varphi(A)$ is the shadow produced by a set A of sources of light.

If $\mathscr{A} = \{A_\lambda \mid \lambda \in \Lambda\}$ is a family of sets, we write

$$\varphi(\mathscr{A}) = \{\varphi\left(A_\lambda\right) \mid \lambda \in \Lambda\}$$

and call $\varphi(\mathscr{A})$ the image of $\mathscr{A}$ under the mapping φ.

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2.17 第 17 页｜ Correspondences

Correspondences

Theorem 3. $A \subseteq A'$ implies that $\varphi(A) \subseteq \varphi(A')$.

Theorem 4. $\varphi\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) = \bigcup_{\lambda\in\Lambda} \varphi(A_\lambda)$.

Theorem 5. $\varphi\left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) \subseteq \bigcap_{\lambda\in\Lambda} \varphi(A_\lambda)$.

Theorem 6. If $A \subseteq S$ and if φ is a mapping of S with range T, then

$$\mathbf{C}(\varphi(A)) \subseteq \varphi(\mathbf{C}A).$$

If, further, φ is injective, then

$$\mathbf{C}(\varphi(A)) = \varphi(\mathbf{C}A).$$

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2.18 第 18 页｜Correspondences

Correspondences

Proof of Theorem 6. $y \in \mathbf{C}(\varphi(A))$, $y \notin \varphi(A) = \bigcup_{x \in A} \varphi(x)$, there isn't $x \in A$ such that $y \in \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$, and hence $y \in \varphi(\mathbf{C}A)$.

Conversely, $y \in \varphi(\mathbf{C}A) = \bigcup_{x \in \mathbf{C}A} \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$. If φ is injective, then $y \notin \varphi(x)$ for $x \in A$, and $y \in \varphi(x)$ for $x \notin A$, thus there isn't $x \in A$ such that $y \in \varphi(x)$, $y \notin \bigcup_{x \in A} \varphi(x) = \varphi(A)$, $y \in \mathbf{C}(\varphi(A))$.

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

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2.19 第 19 页｜ Correspondences

Correspondences

The graph of the correspondence φ is a subset of $S \times T$, namely

$$G(\varphi) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

Example: Consider the case of the rectangle already used to illustrate the graph of a mapping: the graph of correspondence from S to T is visualized as a set of points of the rectangle: the intersection of this set with a vertical through an arbitrary point x of S is a non-empty set (the projection of which on T is $\varphi(x)$).

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

本页重点：函数图像是有序对集合。函数性要求每个 x 有且只有一个 y；对应图像允许同一个 x 对应多个 y。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「graph」证明专题。

2.20 第 20 页｜Correspondences

Correspondences

A binary operation $\top$ on a set S associates with each pair (x, y) of elements of S (the order of which is essential) a unique element z of S. One writes $z = x \top y$. Thus a binary operation on S is nothing else than a function from $S \times S$ to S.

Example: (1) The addition of two positive integers is a binary operation $+$ on $\mathbb{N}$. (2) So is the multiplication of two positive integers. (3) Given two positive integers x, y, the expression x^y note the product of y positive integers equal to x; thus a binary operation $\wedge$ on $\mathbb{N}$ can be defined by $x \wedge y = x^y$.

这一页怎么读

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2.21 第 21 页｜Correspondences

Correspondences

A binary operation $\top$ on S is said to be associative if for all x, y, z in S one has $(x \top y) \top z = x \top (y \top z)$. It is said to be commutative if for all x, y in S one has $x \top y = y \top x$.

Example: (1) Both $+$ and $\cdot$ on $\mathbb{N}$ are associative and commutative. (2) $\wedge$ on $\mathbb{N}$ is neither associative nor commutative.

这一页怎么读

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2.22 第 22 页｜Correspondences

Correspondences

Consider two binary operations $\perp$ and $\top$ defined on S. The first is said to be distributive with respect to the second if for all x, y, z in S one has $x\perp(y\top z) = (x\perp y)\top(x\perp z)$.

Example: (1) On $\mathbb{N}$, $\cdot$ is distributive with respect to $+$. (2) But $+$ is not distributive with respect to $\cdot$.

这一页怎么读

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2.23 第 23 页｜本页课件

Operations on mappings

If φ_1 and φ_2 are two mappings of S into T, their union is the mapping $(\varphi_1 \cup \varphi_2)$ of S into T defined by

$$(\varphi_1 \cup \varphi_2)(x) = \varphi_1(x) \cup \varphi_2(x).$$

The intersection of φ_1 and φ_2 is the mapping $(\varphi_1 \cap \varphi_2)$ of S into T defined by

$$(\varphi_1 \cap \varphi_2)(x) = \varphi_1(x) \cap \varphi_2(x).$$

The Cartesian product of φ_1 and φ_2 is the mapping $(\varphi_1 \times \varphi_2)$ of S into $T \times T$ defined by

$$(\varphi_1 \times \varphi_2)(x) = \varphi_1(x) \times \varphi_2(x).$$

More generally, given any operations $\vee$ and $\wedge$ on sets, we write

$$(\varphi_1 \vee \varphi_2)(x) = \varphi_1(x) \vee \varphi_2(x);$$
$$(\varphi_1 \wedge \varphi_2)(x) = \varphi_1(x) \wedge \varphi_2(x).$$

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

结合原文理解：证明像与交集的关系时，始终检查同一个 y 是否来自同一个 x；逆像的集合运算往往严格保等号。

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2.24 第 24 页｜Operations on mappings

Operations on mappings

If φ_1 is a mapping of S into T and φ_2 is a mapping of T into U, the composition product of φ_2 by φ_1 is the mapping $(\varphi_2 \cdot \varphi_1)$ of S into U defined by

$$(\varphi_2 \cdot \varphi_1)(x) = \varphi_2(\varphi_1(x)).$$

We have

(1) $(\varphi_1 \cup \varphi_2)(A) = \varphi_1(A) \cup \varphi_2(A)$;
(2) $(\varphi_1 \cap \varphi_2)(A) \subseteq \varphi_1(A) \cap \varphi_2(A)$;
(3) $(\varphi_1 \times \varphi_2)(A) \subseteq \varphi_1(A) \times \varphi_2(A)$;
(4) $(\varphi_2 \cdot \varphi_1)(A) = \varphi_2(\varphi_1(A))$.

这一页怎么读

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2.25 第 25 页｜本页课件

Operations on mappings

A mapping Δ is said to be constant if there exists a subset C of T such that $\Delta(x) = C$ for all S. A constant mapping satisfies the following property:

$$(\Delta \cap \varphi)(A) = \Delta(A) \cap \varphi(A).$$

Another important mapping is the identity mapping, which is the single-valued mapping I of S onto S defined by

$$I(x) = \{x\}.$$

$$\begin{aligned}(\Delta \cap \varphi)(A) &= \bigcup_{x \in A} (\Delta \cap \varphi)(x) = \bigcup_{x \in A} [\Delta(x) \cap \varphi(x)] \\ &= \bigcup_{x \in A} [C \cap \varphi(x)] = C \cap \left[\bigcup_{x \in A} \varphi(x) \right] \\ &= C \cap \varphi(A) = \Delta(A) \cap \varphi(A).\end{aligned}$$

这一页怎么读

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2.26 第 26 页｜Operations on mappings If φ1, φ2, φ3 are mappings of S

Operations on mappings

If $\varphi_1, \varphi_2, \varphi_3$ are mappings of S into S, we have associativity of composition product

$$\varphi_3 \cdot (\varphi_2 \cdot \varphi_1)(x) = (\varphi_3 \cdot \varphi_2) \cdot \varphi_1(x).$$

Thus, if φ is a mapping of S into S, we can write

$$\varphi^2(x) = \varphi(\varphi(x))$$
$$\varphi^3(x) = \varphi\left(\varphi^2(x)\right) = \varphi^2(\varphi(x)), \text{ etc.}$$

The transitive closure of φ is a mapping $\hat{\varphi}$ of S into S defined by

$$\hat{\varphi}(x) = \{x\} \cup \varphi(x) \cup \varphi^2(x) \cup \varphi^3(x) \cup \cdots$$

The correspondence $A \longrightarrow \hat{\varphi}(A)$ is a closure operation, for

(1) $\hat{\varphi}(A) \supseteq A$,
(2) $A \supseteq A' \Longrightarrow \hat{\varphi}(A) \supseteq \hat{\varphi}(A')$,
(3) $\hat{\varphi}(\hat{\varphi}(A)) = \hat{\varphi}(A)$.

这一页怎么读

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2.27 第 27 页｜Operations on mappings

Operations on mappings

Theorem 7. If Γ_1 and Γ_2 are two semi-single-valued mappings, $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are semi-single-valued mappings.

Theorem 8. If one of the mappings Γ_1 and Γ_2 is injective, the mappings $\Gamma_1 \cap \Gamma_2$ and $\Gamma_1 \times \Gamma_2$ are injective.

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

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2.28 第 28 页｜ Upper and Lower Inverses If φ is a mapping of S into T,

Upper and Lower Inverses

If φ is a mapping of S into T, its lower inverse is the mapping φ^- of T into S defined by

$$\varphi^-(y) = \{x \in S \mid y \in \varphi(x)\}.$$

This is a mapping whose domain is T^* and whose range is S^*.

Theorem 9. If φ is single-valued, φ^- is injective; if φ is injective, φ^- is single-valued; if φ is semi-single-valued, φ^- is semi-single-valued.

Proof. For $y \neq y'$ we want to prove $\varphi^-(y) \cap \varphi^-(y') = \varnothing$. Otherwise, there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is single-valued, then $y = \varphi(x) = y'$, which is a contradiction.

If $\varphi^-(y) \cap \varphi^-(y') \neq \varnothing$, then there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is injective, then $y = y'$ and hence $\varphi^-(y) = \varphi^-(y')$.

这一页怎么读

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2.29 第 29 页｜Upper and Lower Inverses

Upper and Lower Inverses

For any non-empty subset B of T we have

$$\varphi^{-}(B)=\{x\in S\mid \varphi(x)\cap B\neq\varnothing\}$$

and we also write $\varphi^{-}(\varnothing)=\varnothing$. It is clear that the inverse of φ^{-} is $(\varphi^{-})^{-}=\varphi$ and that $y\in\varphi(x)$ is equivalent to $x\in\varphi^{-}(y)$.

$$\begin{aligned}\left(\varphi^{-}\right)^{-}(A) &= \{y\in T\mid \varphi^{-}(y)\cap A\neq\varnothing\}\\ &= \{y\in T\mid \{x\in S\mid y\in\varphi(x)\}\cap A\neq\varnothing\}\\ &= \{y\in T\mid \{x\in A\mid y\in\varphi(x)\}\neq\varnothing\}\\ &= \{y\in T\mid y\in\varphi(A)\}=\varphi(A)\end{aligned}$$

$$\left(\varphi^{-}\right)^{-}(x)=\{y\in T\mid x\in\varphi^{-}(y)\}=\{y\in T\mid y\in\varphi(x)\}=\varphi(x)$$

这一页怎么读

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2.30 第 30 页｜Upper and Lower Inverses

Upper and Lower Inverses

Apart from the mapping φ^{-}, it is sometimes useful to consider another correspondence of a similar kind, which we call the upper inverse of φ; in this mapping to each subset B of T there corresponds the set

$$\varphi^{+}(B) = \{x \in S^{*} \mid \varphi(x) \subseteq B\}.$$

In particular, if $B = \varnothing$, we have $\varphi^{+}(\varnothing) = \varnothing$. We observe that for all subsets B we have $\varphi^{+}(B) \subseteq \varphi^{-}(B)$.

这一页怎么读

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2.31 第 31 页｜ Upper and Lower Inverses Theorem 10. If B1 and B2 are s

Upper and Lower Inverses

Theorem 10. If B_1 and B_2 are subsets of T, then we have

(1) $\mathbf{C}\varphi^+(B) = \varphi^-(\mathbf{C}B)$ and $\mathbf{C}\varphi^-(B) = \varphi^+(\mathbf{C}B)$.

(2) $\varphi^+(B_1) \cup \varphi^+(B_2) \subseteq \varphi^+(B_1 \cup B_2)$.

(3) $\varphi^-(B_1) \cup \varphi^-(B_2) = \varphi^-(B_1 \cup B_2)$.

Proof.

$$\begin{aligned} \mathbf{C}\varphi^+(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \subseteq B\} \\ &= \mathbf{C}\{x \in S \mid \varphi(x) \cap \mathbf{C}B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap \mathbf{C}B \neq \varnothing\} = \varphi^-(\mathbf{C}B) \end{aligned}$$

$$\begin{aligned} \mathbf{C}\varphi^-(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \cap B \neq \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}B\} = \varphi^+(\mathbf{C}B) \end{aligned}$$

这一页怎么读

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2.32 第 32 页｜ Upper and Lower Inverses

Upper and Lower Inverses

$$
\begin{aligned}
& \varphi^{+}(B_1) \cup \varphi^{+}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \subseteq B_1\} \cup \{x \in S \mid \varphi(x) \subseteq B_2\} \\
\subseteq \; & \{x \in S \mid \varphi(x) \subseteq B_1 \cup B_2\} \\
= \; & \varphi^{+}(B_1 \cup B_2)
\end{aligned}
$$

$$
\begin{aligned}
& \varphi^{-}(B_1) \cup \varphi^{-}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \cap B_1 \neq \varnothing\} \cup \{x \in S \mid \varphi(x) \cap B_2 \neq \varnothing\} \\
= \; & \{x \in S \mid (\varphi(x) \cap B_1) \cup (\varphi(x) \cap B_2) \neq \varnothing\} \\
= \; & \{x \in S \mid \varphi(x) \cap (B_1 \cup B_2) \neq \varnothing\} \\
= \; & \varphi^{-}(B_1 \cup B_2)
\end{aligned}
$$

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2.33 第 33 页｜ Countable Sets

Countable Sets

A sequence $\left(x^1, x^2, \cdots, x^q, \cdots\right)$ of elements of a set S is an intuitive concept. Precisely, it is defined as a function from $\mathbb{N}$ to S. It will he denoted by (x^q).

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2.34 第 34 页｜Countable Sets

Countable Sets

Intuitively, a set S is countable if it has at most as many elements as $\mathbb{N}$. In a precise fashion, a set S is defined as countable if it can be put in one-to-one correspondence with a subset of $\mathbb{N}$. When the countable set S is not empty one can always choose the corresponding subset of $\mathbb{N}$ so that it owns 1, and so that, whenever it owns two positive integers, it owns every positive integer between them. The image of $x \in S$ in the correspondence is then called the rank of x. Thus a set is countable if and only if all its elements can be ranked, no two different elements having the same rank. One can prove that

Theorem 11. The product of finite countable sets is countable.

Example: (1) $\mathbb{N} \times \mathbb{N}$ is countable. (2) However, one can prove that the set of subsets $\mathbb{N}$ of is not countable.

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

本页重点：“像”先选 x 再映到 y；“逆像”先限定 y 的目标集合，再找所有满足 f(x) B 的 x。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「image」证明专题。

2.35 第 35 页｜Countable Sets

Countable Sets

Theorem 12. The set of subsets $\mathbb{N}$ of is not countable.

Proof: We show that there is no surjection from a set A to its power set 2^A. Consider any function $f : A \to 2^A$ and let

$$B = \{a \in A \mid a \notin f(a)\}$$

We claim that there is no $b \in A$ such that $f(b) = B$. Indeed, assume $f(b) = B$ for some $b \in A$. Then either $b \in B$ hence $b \notin f(b)$ which is a contradiction, or $b \notin B$ implying that $b \in B$ which is again a contradiction. Hence the map f is not surjective as claimed. □

这一页怎么读

概念主线：函数把每一个自变量对应到唯一的值；对应允许一个自变量对应一组值。值域、像、逆像的运算规则并不对称。

结合原文理解：证明像与交集的关系时，始终检查同一个 y 是否来自同一个 x；逆像的集合运算往往严格保等号。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：graph

证明｜函数图像为何能还原函数

给定 $f:S\to T$，图像 $G(f)=\{(x,f(x)):x\in S\}$。对每个 $x\in S$，图像中至少有 $(x,f(x))$，所以存在性成立；若 $(x,y_1),(x,y_2)\in G(f)$，则 $y_1=f(x)=y_2$，所以唯一性成立。反过来，一个 $G\subseteq S\times T$ 若对每个 x 都存在唯一 y 使 $(x,y)\in G$，即可定义函数 $f(x)=y$。

3.2 专题：image

证明｜像保持并集，但不一定保持交集

任取族 $A_i\subseteq S$。由定义 $y\in f(\bigcup_i A_i)\Leftrightarrow\exists x\in\bigcup_i A_i:f(x)=y\Leftrightarrow\exists i\,\exists x\in A_i:f(x)=y\Leftrightarrow y\in\bigcup_i f(A_i)$。对交集只有 $f(\bigcap_i A_i)\subseteq\bigcap_i f(A_i)$，因为右边的不同集合可能分别用不同的原像点表示同一个 y。反例：$f(1)=f(2)=0$，$A=\{1\}$，$B=\{2\}$，则 $f(A\cap B)=\varnothing$，而 $f(A)\cap f(B)=\{0\}$。对于逆像，$f^{-1}(\bigcap_i B_i)=\bigcap_i f^{-1}(B_i)$ 总成立。

3.3 专题：uncountable

证明｜ Cantor 对角线法：$[0,1]$ 不可数

反设 $(0,1)$ 可以列为 $x_1,x_2,\ldots$。统一选用不以无穷多个 9 结尾的十进制展开。构造新小数 $y=0.y_1y_2\cdots$，其中若 x_n 的第 n 位是 1 就选 $y_n=2$，否则选 $y_n=1$。这样 $y\in(0,1)$，而且对每个 n，y 的第 n 位与 x_n 不同，故 $y\neq x_n$。这与列举已包含全部实数矛盾，所以 $(0,1)$ 不可数。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?